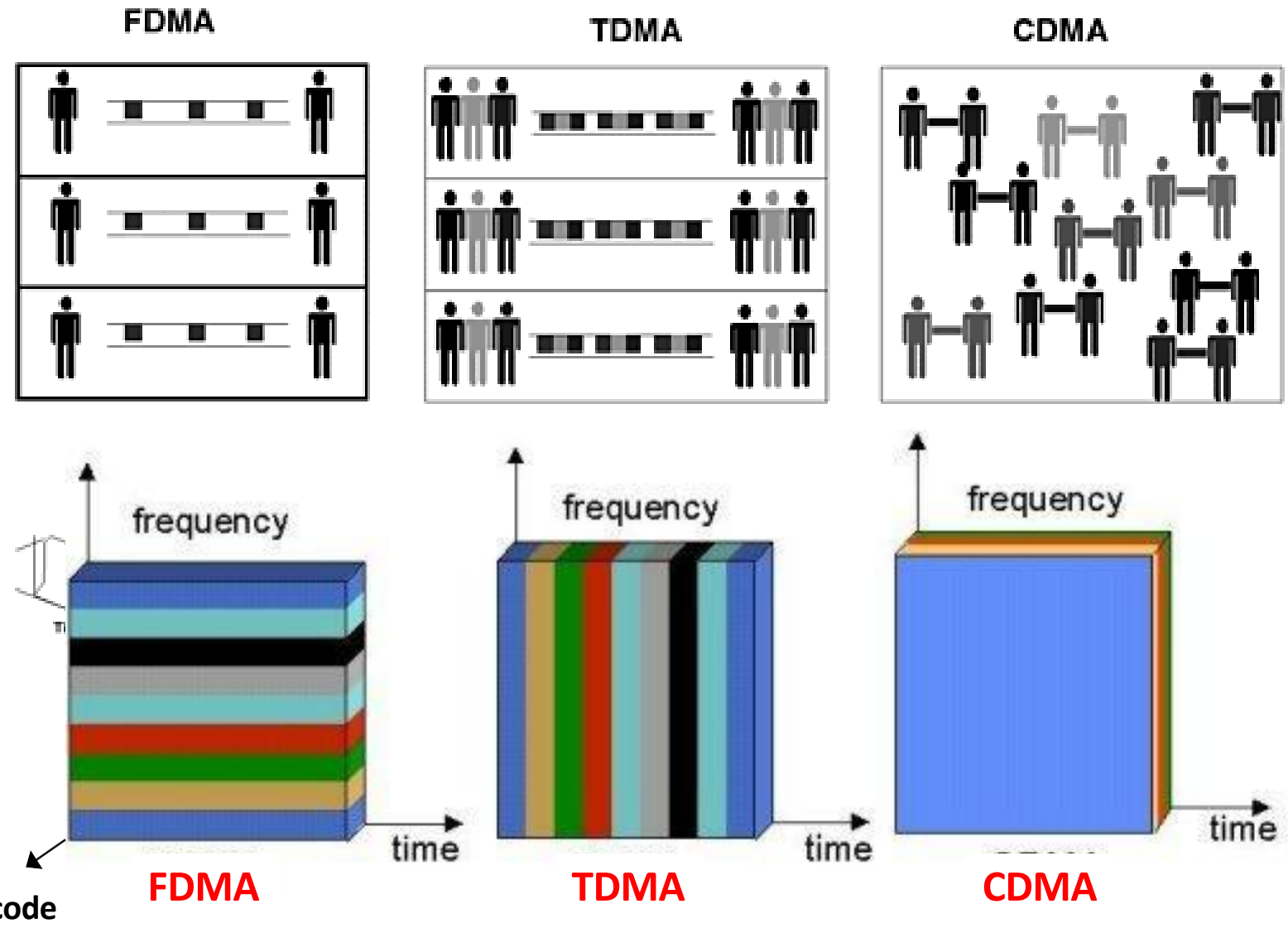
Telecommunication Systems
ICE 3109

Lecture 11
Multiple Access Techniques

Multiple Access Techniques

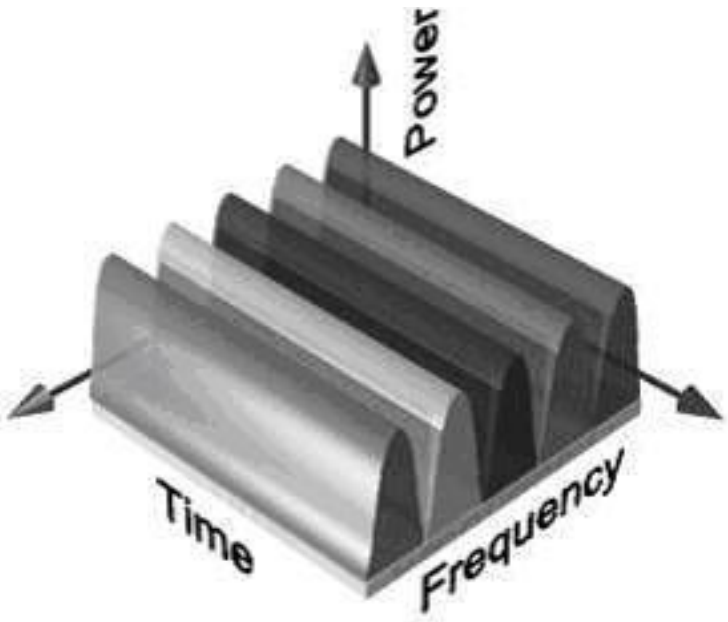
- **Multiple Access** is the use of multiplexing techniques to provide communication service to multiple users over a single channel.
- It allows for many users at one time by sharing a finite amount of spectrum.

Multiple Access Options: Frequency, Time or Code



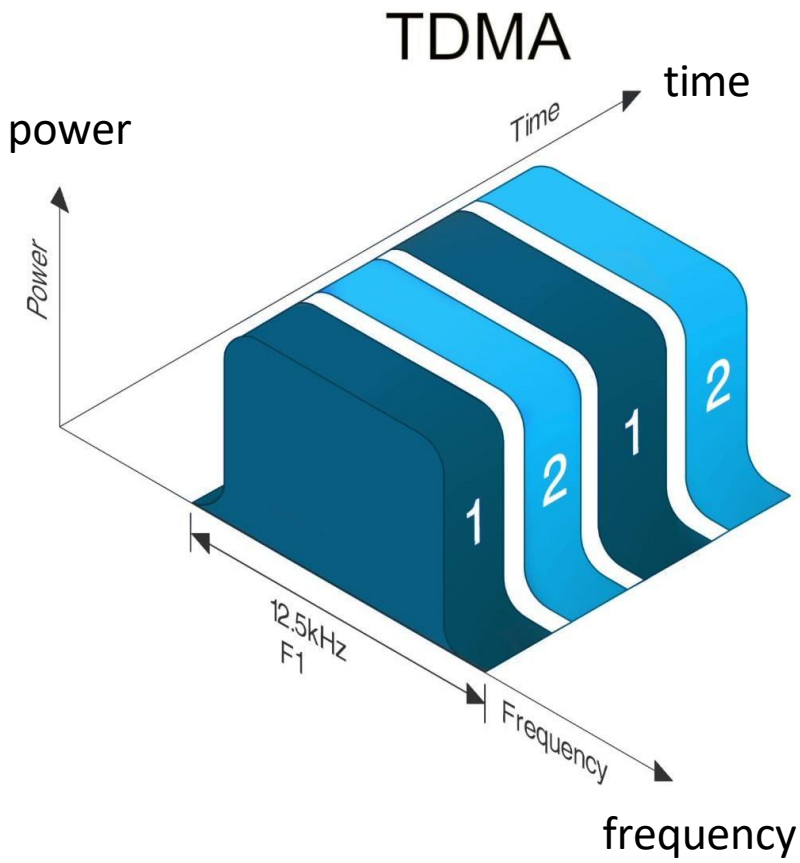
Frequency Division Multiple Access

The available bandwidth is subdivided into a number of narrower band channels. Each user is allocated a unique frequency band in which to transmit and receive on.



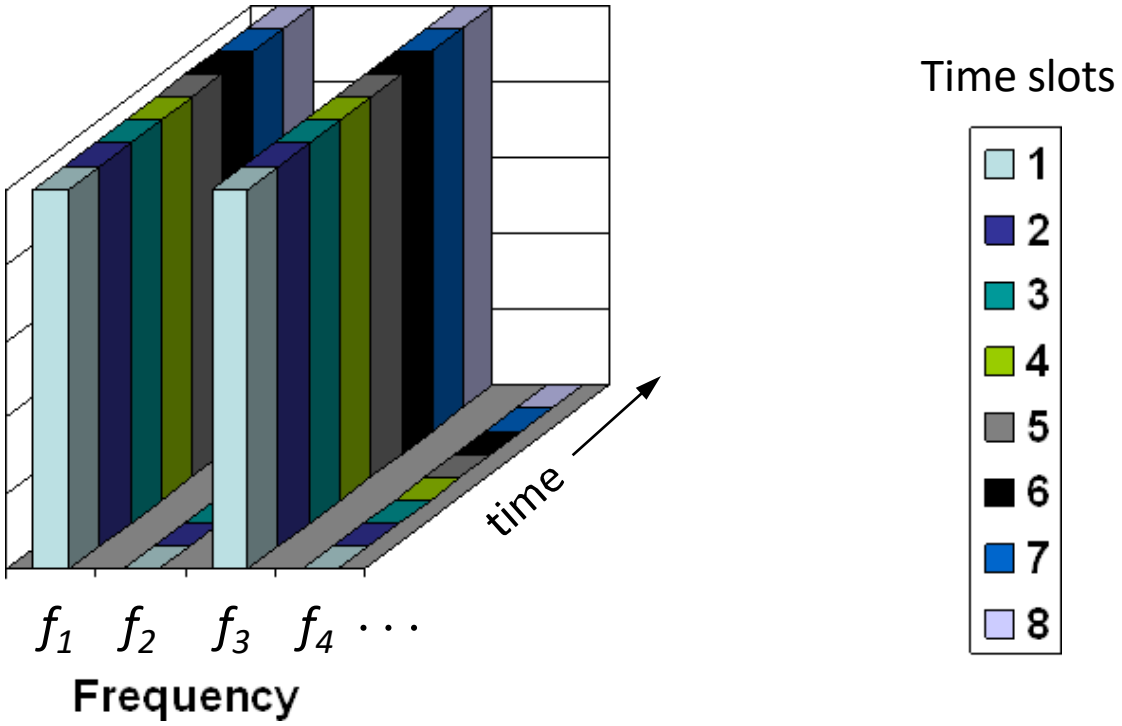
Wavelength Division Multiple Access (WDMA) is a related to FDMA. Wavelength is used in optical fiber communication systems to partition channels. All wavelengths propagate within a single optical fiber.

Time Division Multiple Access



Two time slots are shown

GSM Cellular Uses Both FDMA and TDMA

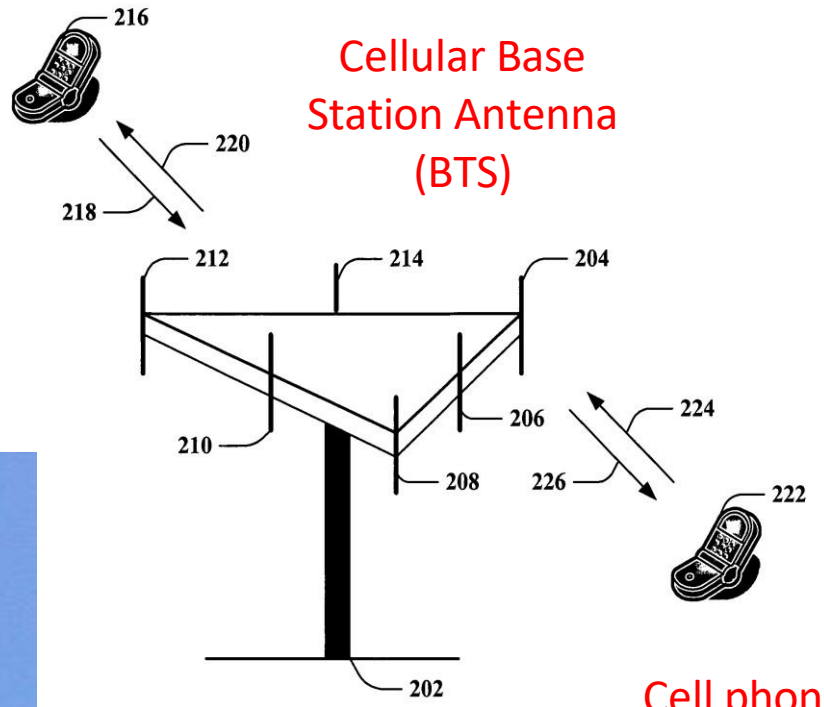


GSM uses a combination of both TDMA and FDMA techniques. The FDMA element divides the assigned frequency of **25 MHz bandwidth** into **124 carrier frequencies**, all spaced 200 kHz apart. The carriers are also divided in time using TDMA. Different users of each RF channel are allocated different time slots (there are 8 time slots per channel).

Spatial Division Multiple Access

EXAMPLE: Cellular telephony uses sectored base station antenna.

Cell phone (MS)



Cellular Base Station Antenna (BTS)

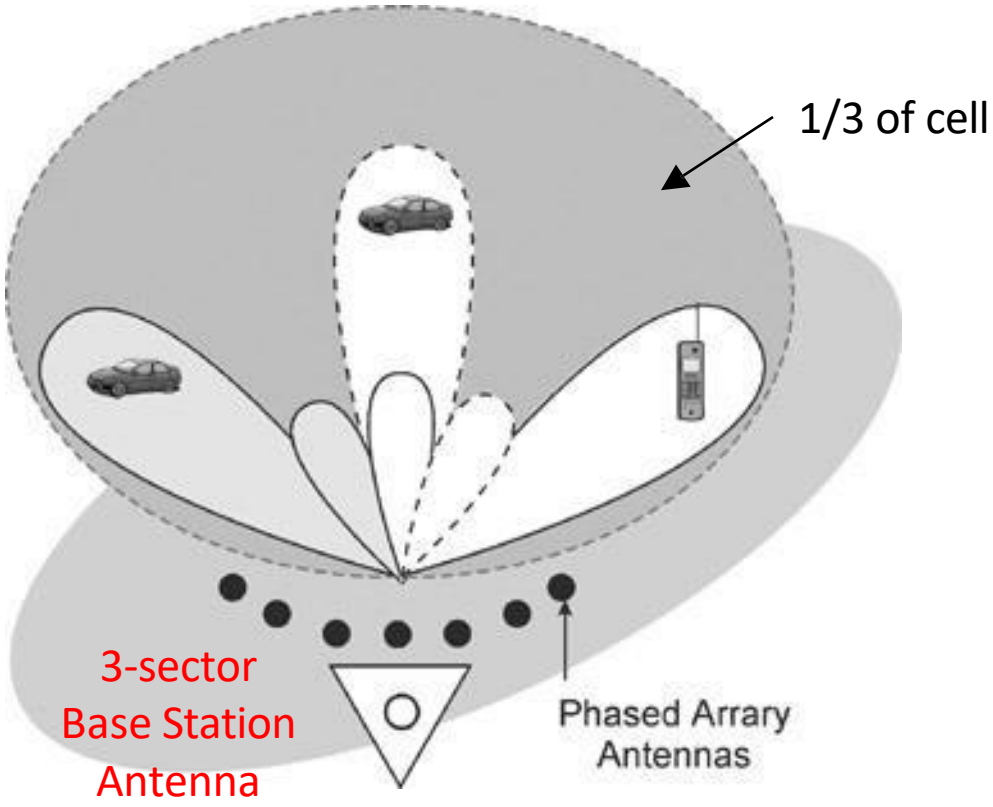


Cell phone (MS)

http://www.google.ms/patents/US8363603

Beam Division Multiple Access (BDMA)

Also known as "**Spatial Division Multiple Access**" (SDMA)



BDMA uses multiple radiators to form multiple beams simultaneously in a mobile telephone cell, thus, providing for multiple access.

Beam Division Multiple Access (BDMA) in Satellite Systems

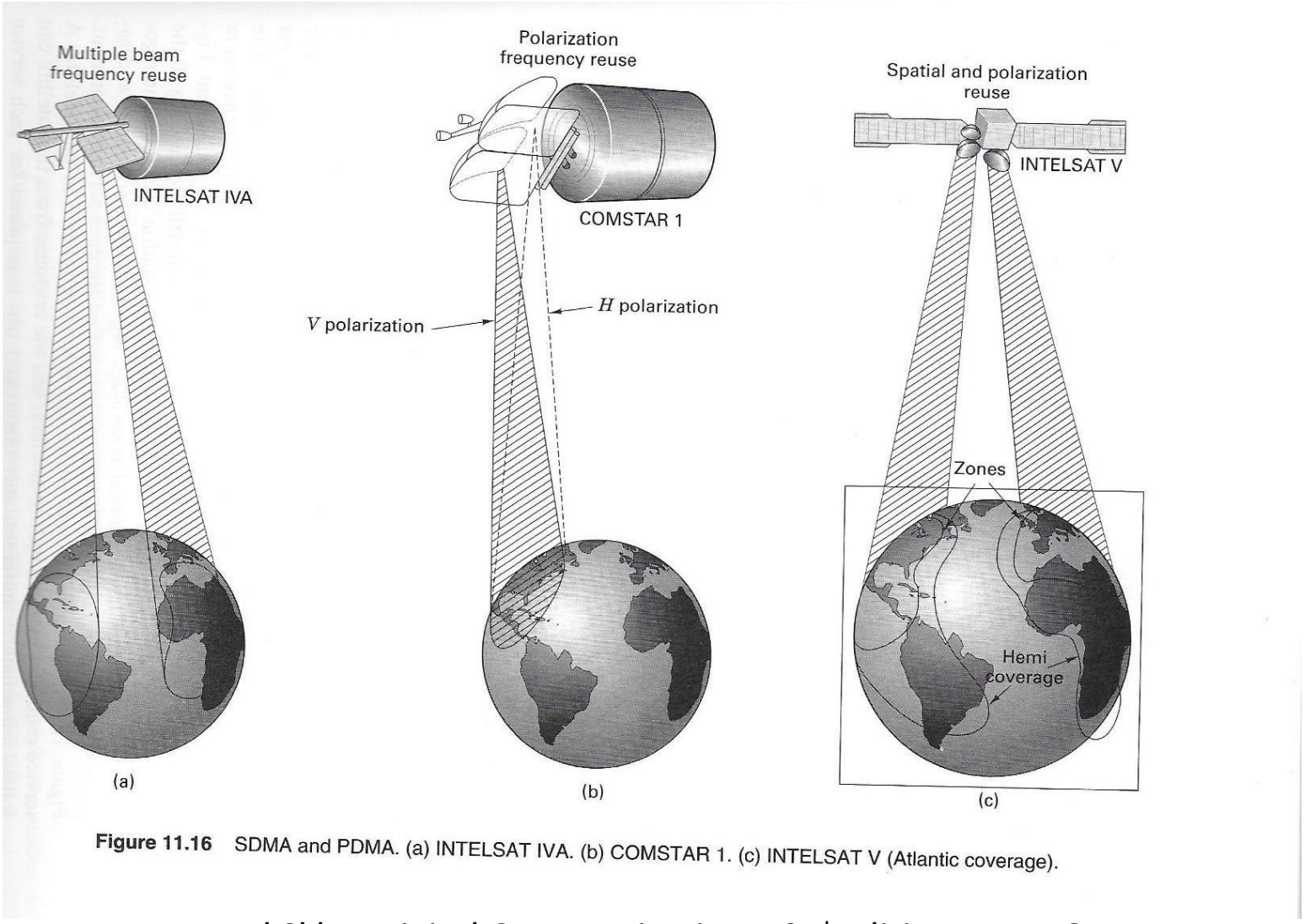


Figure 11.16 SDMA and PDMA. (a) INTELSAT IVA. (b) COMSTAR 1. (c) INTELSAT V (Atlantic coverage).

From Bernard Sklar, Digital Communications, 2nd edition, page 675.

Code Division Multiple Access (CDMA) – I

The CDMA standard was originally designed by Qualcomm in the U.S. and is primarily used in the U.S. and portions of Asia by other carriers.

Salient Features of CDMA:

CDMA is based on the spread spectrum technique

1. In CDMA, every channel uses the full available spectrum.
2. Individual conversations are encoded with a pseudo-random digital sequence and then transmitted.
3. CDMA consistently provides better capacity for voice and data communications, allowing more subscribers to connect at any given time.

CDMA is the common platform on which 3G technologies are built. It is used by Verizon, and Sprint .

A duplex method whereby the Uplink and the Downlink transmissions use two separate frequency bands –

Uplink – 1920 MHz to 1980 MHz

Downlink – 2110 MHz to 2170 MHz

Bandwidth – Each carrier located at center in 5 MHz band

Code Division Multiple Access (CDMA) – II

CDMA allows up to 61 concurrent users in a 1.2288 MHz channel by processing each voice packet with its PN code. There are 64 Walsh codes available to differentiate between calls. Operational limits and quality issues will reduce the maximum number of calls somewhat lower than this value.

In fact, many different "signals" baseband with different spreading codes can be modulated on the same carrier to allow many different users to be supported. Using different orthogonal codes, interference between the signals is minimal. Conversely, when signals are received from several mobile stations, the base station is capable of isolating each because they have different orthogonal spreading codes.

